EX. NO: 7

IMPLEMENTATION OF RSA

AIM:

To write a C program to implement the RSA encryption algorithm.

ALGORITHM:

STEP-1: Select two co-prime numbers as p and q.

STEP-2: Compute n as the product of p and q.

STEP-3: Compute (p-1)*(q-1) and store it in z.

STEP-4: Select a random prime number e that is less than that of z.

STEP-5: Compute the private key, d as e * mod-1 (z).

STEP-6: The cipher text is computed as messagee * mod n. STEP-7: Decryption is done as cipherdmod n.

PROGRAM:

```
#include <stdio.h>


long long modExp(long long base, long long exp, long long mod)
{
   long long result = 1;
   base = base % mod;


   while (exp > 0)
   {
      if (exp % 2 == 1)
         result = (result * base) % mod;


      exp = exp / 2;
      base = (base * base) % mod;
   }
```

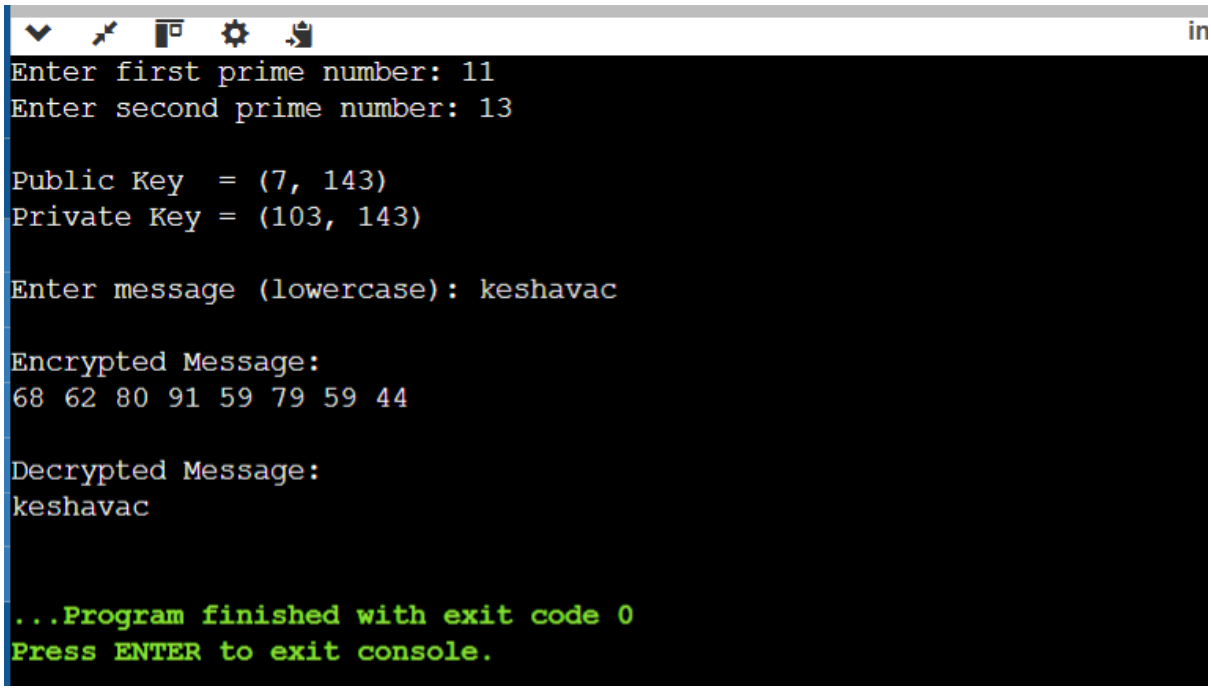
```
    return result;
}

int gcd(int a, int b)
{
    while (b != 0)
    {
        int t = b;
        b = a % b;
        a = t;
    }
    return a;
}

int modInverse(int e, int phi)
{
    int d;

    for (d = 1; d < phi; d++)
    {
        if ((e * d) % phi == 1)
            return d;
    }

    return -1;
}

int isPrime(int n)
{
```

```
    if (n < 2)
        return 0;

    for (int i = 2; i * i <= n; i++)
    {
        if (n % i == 0)
            return 0;
    }

    return 1;
}

int main()
{
    int p, q;

    printf("Enter first prime number: ");
    scanf("%d", &p);

    printf("Enter second prime number: ");
    scanf("%d", &q);

    if (!isPrime(p) || !isPrime(q) || p == q)
    {
        printf("Invalid prime numbers.\n");
        return 0;
    }

    int n = p * q;
```

```
int phi = (p - 1) * (q - 1);

int e;

for (e = 2; e < phi; e++)
{
   if (gcd(e, phi) == 1)
      break;
}

int d = modInverse(e, phi);

printf("\nPublic Key  = (%d, %d)\n", e, n);
printf("Private Key = (%d, %d)\n", d, n);

char msg[100];

printf("\nEnter message (lowercase): ");
scanf("%s", msg);

long long cipher[100];

printf("\nEncrypted Message:\n");

int i;

for (i = 0; msg[i] != '\0'; i++)
{
   int m = msg[i];
```

```
        cipher[i] = modExp(m, e, n);
        printf("%lld ", cipher[i]);
    }

    printf("\n\nDecrypted Message:\n");

    for (i = 0; msg[i] != '\0'; i++)
    {
        char ch = (char)modExp(cipher[i], d, n);
        printf("%c", ch);
    }

    printf("\n");

    return 0;
}
```

OUTPUT:



RESULT:

Thus the C program to implement RSA encryption technique had been implemented successfully